WORKED SOLUTION

STEP II 2025, Question 8

Pure • Balanced Partitions of Integers

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

The key idea

Shifting every element of a set by the same amount c changes its sum and its sum of squares in a controlled way: the change depends only on how many elements there are, their sum and (for squares) nothing more. So if two sets of the same size already agree in their sums and sums of squares, shifting one copy of each and swapping them over builds a bigger balanced partition. Part (ii) is the algebra behind this, and parts (iii) and (iv) are two uses of it.

Solution

Part (i)

The total of $1, 2, \ldots, 8$ is 36 and the total of their squares is 204, so each half needs sum 18 and sum of squares 102.

Put 8 in B. The other three elements of B must have sum 10 and squares summing to $102 - 64 = 38$. The triples of distinct numbers from 1 to 7 with sum 10 are

$$\{1,2,7\},\ \{1,3,6\},\ \{1,4,5\},\ \{2,3,5\},$$

with sums of squares 54, 46, 42 and 38. Only $\{2,3,5\}$ works, so

$$A = \{1,4,6,7\}, \qquad B = \{2,3,5,8\}.$$

Check: $1+4+6+7 = 18 = 2+3+5+8$ and $1+16+36+49 = 102 = 4+9+25+64$. (The search also shows this is the only balanced partition into fours, apart from swapping A and B.)

Part (ii)

Expand each cube with the binomial theorem:

$$\sum_{k=1}^{m} (c+b_k)^3 = mc^3 + 3c^2 \sum_{k=1}^{m} b_k + 3c \sum_{k=1}^{m} b_k^2 + \sum_{k=1}^{m} b_k^3,$$

and similarly with a_k in place of b_k. Hence

$$\left(\sum a_k^3 + \sum (c+b_k)^3\right) - \left(\sum b_k^3 + \sum (c+a_k)^3\right)$$
$$= 3c^2\left(\sum b_k - \sum a_k\right) + 3c\left(\sum b_k^2 - \sum a_k^2\right),$$

because the mc^3 terms cancel, as do the two copies of $\sum a_k^3$ and of $\sum b_k^3$. Both brackets are zero by the hypotheses, so the two sides are equal for every real c, as required.

Notice that the same expansion one power lower gives

$$\sum a_k^2 + \sum (c+b_k)^2 - \sum b_k^2 - \sum (c+a_k)^2 = 2c\left(\sum b_k - \sum a_k\right) = 0,$$

and with first powers $\sum a_k + \sum (c+b_k) = \sum b_k + \sum (c+a_k)$, since both equal $mc + \sum a_k + \sum b_k$.

Part (iii)

Take the balanced partition from (i), with a_k running through $1, 4, 6, 7$ and b_k through $2, 3, 5, 8$, and take $c = 8$. Then $c + a_k$ runs through $9, 12, 14, 15$ and $c + b_k$ through $10, 11, 13, 16$. Set

$$A = \{1, 4, 6, 7\} \cup \{10, 11, 13, 16\}, \qquad B = \{2, 3, 5, 8\} \cup \{9, 12, 14, 15\}.$$

These are disjoint, each has 8 elements, and together they are $\{1, 2, \ldots, 16\}$, because $\{1, \ldots, 8\}$ and $\{9, \ldots, 16\}$ are each split between them.

> **Ask yourself** Why put the shifted copy of B into A, rather than the shifted copy of A?

Because that is exactly the pattern in (ii): the left-hand side there is $\sum a_k^3 + \sum (c+b_k)^3$, which is the sum of cubes over our A, and the right-hand side is the sum of cubes over our B. So (ii) with $c = 8$ gives equal sums of cubes. The two extra identities at the end of (ii) give equal sums and equal sums of squares, so the partition is balanced too.

Directly: both halves have sum 68, sum of squares 748 and sum of cubes 9248.

Part (iv)

Write the elements of S as $(n+k)^2$ for $k = 0, 1, \ldots, 11$. For any set K of values of k,

$$\sum_{k \in K} (n+k)^2 = |K|\, n^2 + 2n \sum_{k \in K} k + \sum_{k \in K} k^2.$$

So if we split $\{0, 1, \ldots, 11\}$ into two sets that have the same *size*, the same sum and the same sum of squares, the corresponding squares will have equal totals for every n.

The given partition of $\{1, \ldots, 11\}$ has equal sums (33 each) and equal sums of squares (253 each), but its parts have sizes 6 and 5. Adding $k = 0$ to the smaller part fixes this: it increases the size by one and adds nothing to the sum or the sum of squares. So we use

$$\{1, 3, 4, 5, 9, 11\} \quad \text{and} \quad \{0, 2, 6, 7, 8, 10\},$$

which have size 6, sum 33 and sum of squares 253 each. Therefore

$$C = \{(n+1)^2, (n+3)^2, (n+4)^2, (n+5)^2, (n+9)^2, (n+11)^2\},$$
$$D = \{n^2, (n+2)^2, (n+6)^2, (n+7)^2, (n+8)^2, (n+10)^2\}$$

both have element sum $6n^2 + 66n + 253$. The twelve squares are distinct since $n > 0$, so C and D are disjoint and their union is S.

Checks

- With $n = 1$ in (iv), $C = \{4, 16, 25, 36, 100, 144\}$ and $D = \{1, 9, 49, 64, 81, 121\}$, both summing to $325 = 6 + 66 + 253$.
- In (iii), the sums 68 and sums of squares 748 are half of 136 and 1496, the totals for 1 to 16; the sums of cubes are 9248, half of $136^2 = 18496$.
- A computer search over all ways of choosing four of $1, \ldots, 8$ confirms the partition in (i) is unique up to swapping, and the construction in (iv) was checked for n up to 499.

Common slips

- In (iv), using the given A and B directly: the sizes 6 and 5 differ, so the n^2 terms do not cancel. The missing element n^2 (that is, $k = 0$) must go with B.
- In (iii), shifting the wrong half: putting $\{9, 12, 14, 15\}$ (the shifted a_k) with $\{1, 4, 6, 7\}$ still balances sums and squares, but the cube sums then differ by $2\left(\sum b_k^3 - \sum a_k^3\right) = 2(672 - 624) = 96$.

Beyond STEP

This is the Prouhet–Tarry–Escott problem: split a set of integers into two parts whose sums of jth powers agree for $j = 1, \ldots, k$. Repeating the shift-and-swap of (iii) with $c = 16, 32, \ldots$ produces the Thue–Morse partition of $\{0, 1, \ldots, 2^{k+1} - 1\}$, which balances all powers up to k; the proof is the binomial expansion of (ii) done by induction. Davenport's *The Higher Arithmetic* is a good companion for this kind of elementary number theory.

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